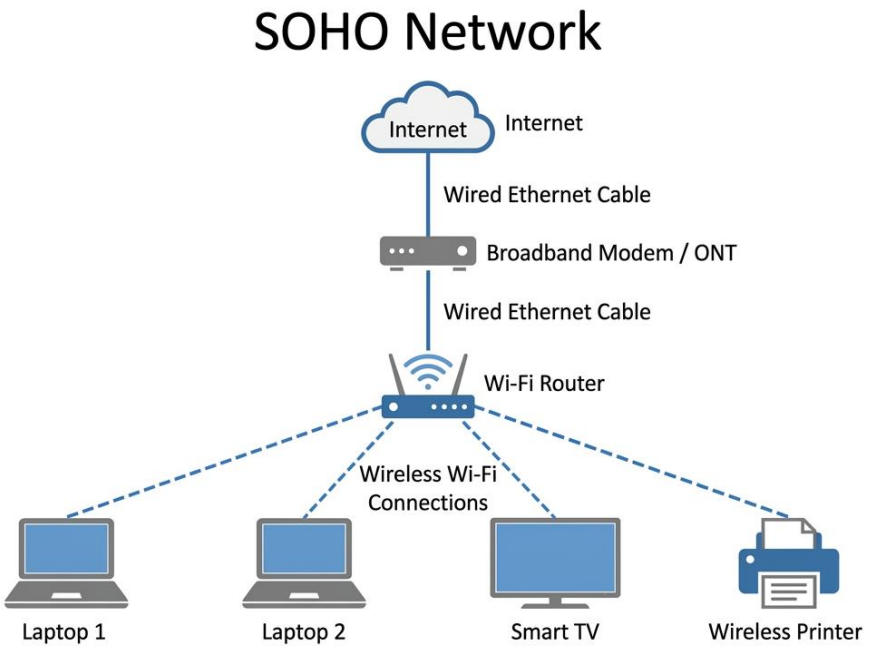
# Build a SOHO Network

## 1. Network Diagram




## Working:

The broadband connection provides internet access. The Wi-Fi router shares the internet connection with two laptops, one smart TV, and one printer through Wi-Fi.

## 2. Hardware Components Required

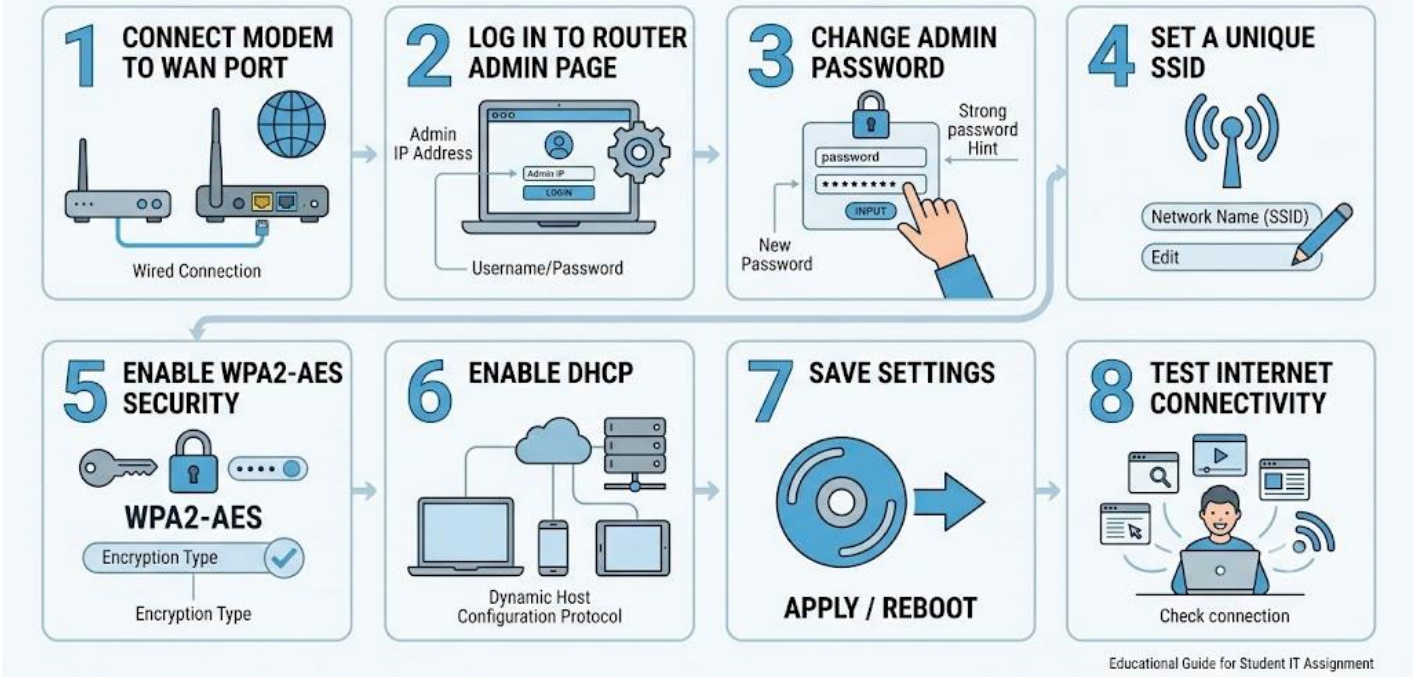
1. Broadband Internet Connection – JioFiber or Airtel Xstream Fiber
2. Wi-Fi Router – TP-Link Archer C6
3. Broadband Modem/ONT – Provided by ISP
4. Two Wi-Fi-enabled laptops
5. Smart TV with Wi-Fi support
6. Wireless Printer – Epson EcoTank L3250
7. Cat6 Ethernet Cable
8. Power Adapters
9. UPS (Optional)

A network switch or Wi-Fi extender is not required because all devices can connect wirelessly to the router.

## 3. Wi-Fi Router Configuration

1. Connect the broadband modem/ONT to the WAN port of the Wi-Fi router using an Ethernet cable.
2. Switch on the modem and router.
3. Connect a laptop to the router using Wi-Fi or an Ethernet cable.
4. Open a web browser and enter the router's management address given in its manual.
5. Log in to the router administration page.
6. Change the default administrator password.
7. Open Wireless Settings and set the SSID to `Dhairya_SOHO_Network`.
8. Select WPA2-PSK/WPA2-Personal security with AES encryption.
9. Set a strong Wi-Fi password.
10. Enable DHCP and configure the address pool from `192.168.1.100` to `192.168.1.199`.
11. Save the settings and reconnect all devices.
12. Test the internet connection on both laptops and the smart TV.

# STEPS TO CONFIGURE A WI-FI ROUTER




Educational Guide for Student IT Assignment

# 4. IP Address Configuration

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

The following MAC addresses are fictional examples for this assignment.

| Device | MAC Address | IP Address |
|---|---|---|
| Wi-Fi Router | 02:11:22:33:44:01 | 192.168.1.1 |
| Laptop 1 | 02:11:22:33:44:02 | 192.168.1.100 (DHCP) |
| Laptop 2 | 02:11:22:33:44:03 | 192.168.1.101 (DHCP) |
| Smart TV | 02:11:22:33:44:04 | 192.168.1.20 (Static) |
| Printer | 02:11:22:33:44:05 | 192.168.1.30 (Static) |

## Configuration Steps

1. Log in to the router administration page.

2. Enable DHCP for the laptops.
3. Assign a static IP address of `192.168.1.20` to the smart TV.
4. Assign a static IP address of `192.168.1.30` to the printer.
5. Set the subnet mask to `255.255.255.0`.
6. Set the default gateway to `192.168.1.1` for the smart TV and printer.
7. Set DNS to `192.168.1.1`, if supported.
8. Check the IP addresses using the `ipconfig` command on the laptops.

## 5. Troubleshooting

### Problem 1: Wi-Fi Is Disabled or Password Is Incorrect

Diagnosis: Check whether Wi-Fi is enabled and whether the correct SSID is visible.

Solution: Enable Wi-Fi, select the correct network, enter the correct password and reconnect.

### Problem 2: DHCP or IP Address Problem

Diagnosis: Open Command Prompt and run `ipconfig`. Check whether the laptop has received a valid IP address.

Solution: Set IP configuration to automatic and run `ipconfig /release` followed by `ipconfig /renew`. Ensure DHCP is enabled on the router.

### Problem 3: Router or Internet Connection Problem

Diagnosis: Check the router indicator lights and test the internet on another device.

Solution: Check the Ethernet cable between the modem and router. Restart the modem and router if necessary. Contact the ISP if the internet service is down.

### Problem 4: DNS Problem

Diagnosis: Run `ping 192.168.1.1` and `ping 8.8.8.8` in Command Prompt.

Solution: Check DNS settings and run `ipconfig /flushdns`. If the problem continues, check or update the Wi-Fi network adapter.

## Conclusion

A SOHO network allows multiple devices to share one internet connection. Proper router configuration, IP addressing and troubleshooting help maintain a reliable and secure network.